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Group Project Risk and Portfolio Analysis

Project 1

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Contents

1 Problem 1 **1**
1.1 Theoretical introduction 1
1.2 Testing Gaussian IID hypothesis 1

2 Problem 2 **6**
2.1 Theoretical introduction 6
2.2 Computation and observations 8

3 Problem 3 **10**
3.1 GARCH processes 10
3.2 Computation 10

4 Problem 4 **11**

5 Problem 5 **15**

6 Problem 6 **17**

References **19**

Introduction

The aim of this project is to study and investigate the characteristics of the Standard & Poor's Europe 350 index with its daily closing prices from April 2014 to April 2019, comprising 1,252 observations. The primary objective is to evaluate different modeling approaches for the daily scaled log-returns of the index and to determine their impact on risk estimation. By comparing different models' assumptions, this report aims to provide a comprehensive assessment of how best to describe and quantify risk in the S&P Europe 350 index. All the analysis are performed in MATLAB.

1 Problem 1

1.1 Theoretical introduction

The first analysis aims to determine whether the scaled log-returns, defined as

$$Y_t = a\big(\log V_t - \log V_{t-1}\big), \qquad \text{with } a = 100,$$

can reasonably be assumed to satisfy

$$Y_t \sim \mathcal{N}(\mu, \sigma^2), \quad \text{and IID.}$$

First, recall that if the series $\{Y_t\}$ is **IID Gaussian** it means that it has:

- Independence: past returns do not influence future returns;
- Identical distribution: the mean and variance of returns are constant over time;
- Normality: each return follows a Gaussian (normal) distribution.

Secondly, let us introduce the concept of **volatility**, which will be widely used throughout this report. Volatility is a statistical measure of the variability of a market index's returns; in other words, it provides a quantification of risk. Assets with higher volatility are generally considered riskier than those with lower volatility, as their prices are less predictable.

Mathematically, volatility is defined as the standard deviation of returns.

Under the IID assumption, volatility is constant and can be expressed as

$$\sigma = \sqrt{\mathrm{Var}(Y_t)}.$$

Intuitively, this implies that the level of risk today is the same as it was yesterday and will be tomorrow.

1.2 Testing Gaussian IID hypothesis

We performed the analysis on the data to test our hypothesis using different methods. We will now present each of these tests, showing the results obtained.

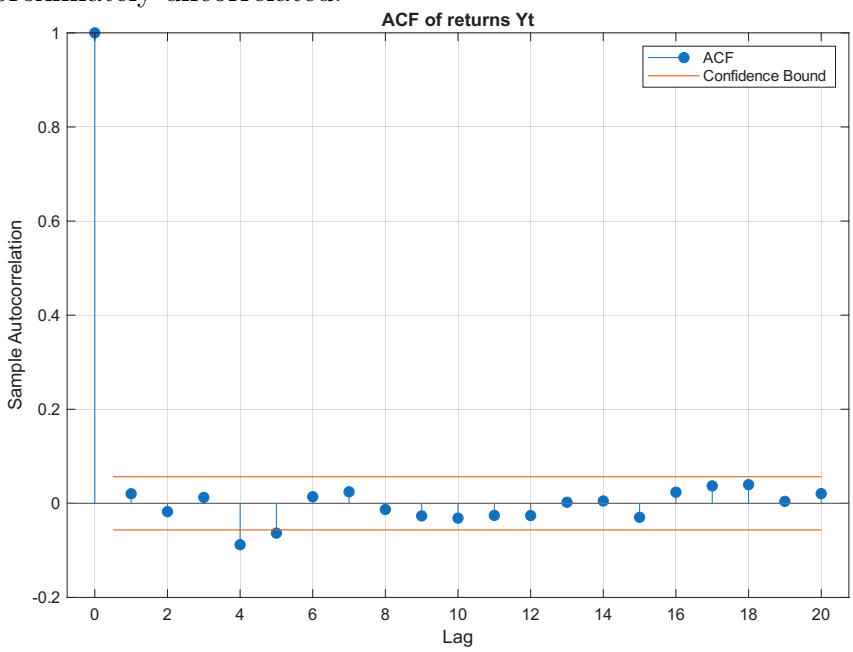
ACF of returns

Autocorrelation measures how a time series is correlated with its own past. In particular, the autocorrelation function (ACF) measures linear dependence between observations at different lags. For lag k, it is defined as:

$$\rho(k) = \mathrm{Corr}(Y_t, Y_{t-k}), \qquad \hat{\rho}_k = \frac{\sum_{t=k+1}^{T}(Y_t - \bar{Y})(Y_{t-k} - \bar{Y})}{\sum_{t=1}^{T}(Y_t - \bar{Y})^2}.$$

The MATLAB command `autocorr` plots the autocorrelations and includes 95% confidence bounds. The values lying within these bounds are statistically zero.

The ACF plot for Y_t, with the choice of 30 lags, shows that most autocorrelations lie within the confidence bounds. This means that returns are approximately uncorrelated.



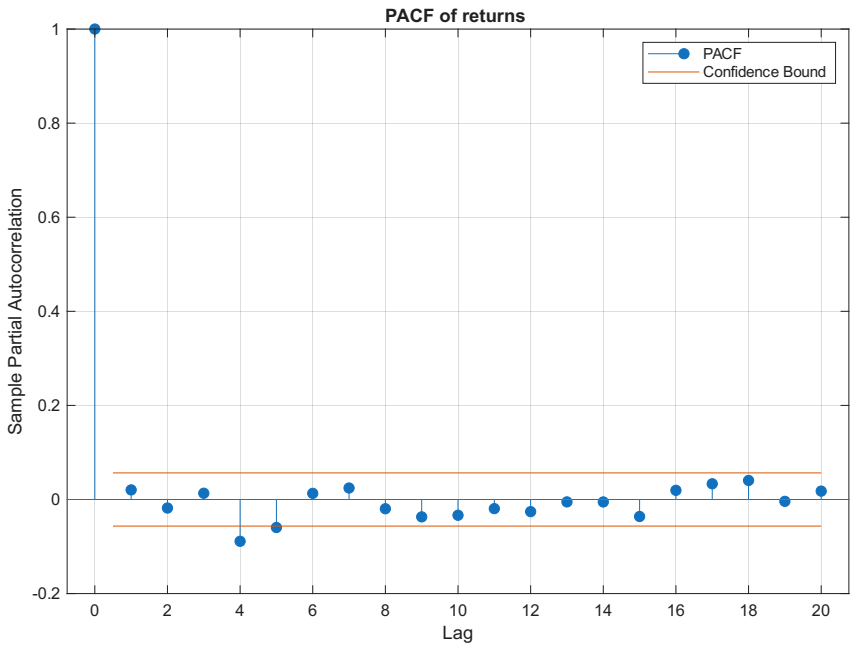
PACF of returns

The partial autocorrelation function (PACF) at lag k is defined as:

$$\mathrm{PACF}(k) = \mathrm{Corr}(Y_t, Y_{t-k} \mid Y_{t-1}, \ldots, Y_{t-k+1}).$$

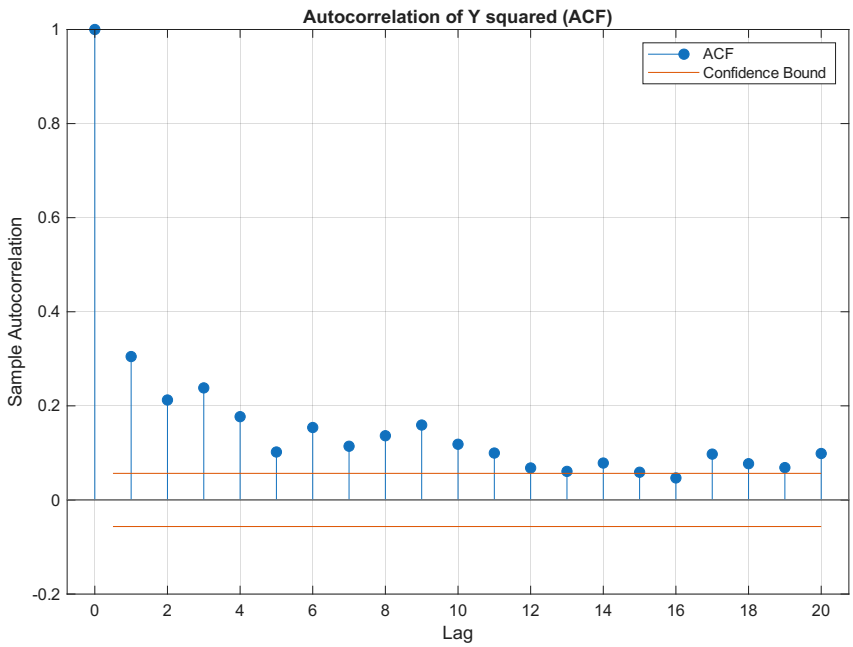
Mathematically, it measures the correlation between Y_t and Y_{t-k} after removing the effects of intermediate lags.

Through MATLAB command `parcorr` we computed the PACF of Y_t for 30 lags and obtained no significant values beyond the confidence bounds. Therefore, it is possible to derive the same conclusions as before: there is no significant correlation between returns.



ACF of squared returns

When we analyze the autocorrelation function of squared returns Y_t^2, we observe completely different behavior compared to the returns. Several autocorrelation values lie outside the confidence bounds, so they are statistically significant.



This observation is fundamental from a financial perspective. Since the mean return μ is typically close to zero, we have:

$$\mathbb{E}[Y_t^2] = \mathrm{Var}(Y_t) + \mu^2 \approx \mathrm{Var}(Y_t),$$

and therefore squared returns provide a useful approximation of variance. As volatility is defined as the standard deviation of returns,

$$\sigma_t = \sqrt{\mathrm{Var}(Y_t)},$$

analyzing Y_t^2 allows us to investigate the dynamics of volatility, in particular to detect volatility clustering, which is the tendency of volatility to persist over time: large movements are often followed by large movements, and small movements by small movements.

The presence of significant autocorrelations provides evidence of volatility clustering. This leads us to conclude that the IID assumption does not hold for our data.

Ljung–Box Test

The Ljung–Box test is a statistical tool used to determine whether a set of autocorrelations is jointly equal to zero. The hypotheses of the test are as follows:

- H_0: No autocorrelation up to the chosen lag;
- H_1: At least one autocorrelation is nonzero.

The command `lbqtest` returns a p-value that needs to be compared with the standard 5% significance level: if the obtained value is higher, we fail to reject the null hypothesis H_0.
Applying to returns Y_t, the null hypothesis is **not rejected**, confirming the absence of linear dependence.

```
--- Ljung-Box on Returns (Lag 15) ---
P-Value: 0.10624
Result: Fail to Reject Null Hypothesis (h=0)
Conclusion: H0 is supported.
```

Applying to squared returns Y_t^2, the null hypothesis is **strongly rejected** confirming that data cannot be considered independent identically distributed.

```
    --- Ljung-Box on Squared Returns (Lag 15) ---
P-Value: 0.00000
Result: Reject Null Hypothesis (h=1)
Conclusion: H0 is not supported
```

Lilliefors Test for Normality

The Lilliefors test is a statistical test that aims to check whether a dataset follows normal distribution, for cases where the mean and variance are estimated from the data. The command `lillietest` returns a p-value that needs to be compared with the standard 5% significance level: if the obtained value is higher, we fail to reject the null hypothesis H_0. The hypotheses of the test are as follows:

- H_0: The datas follow a normal distribution;
- H_1: The datas are not normally distributed.

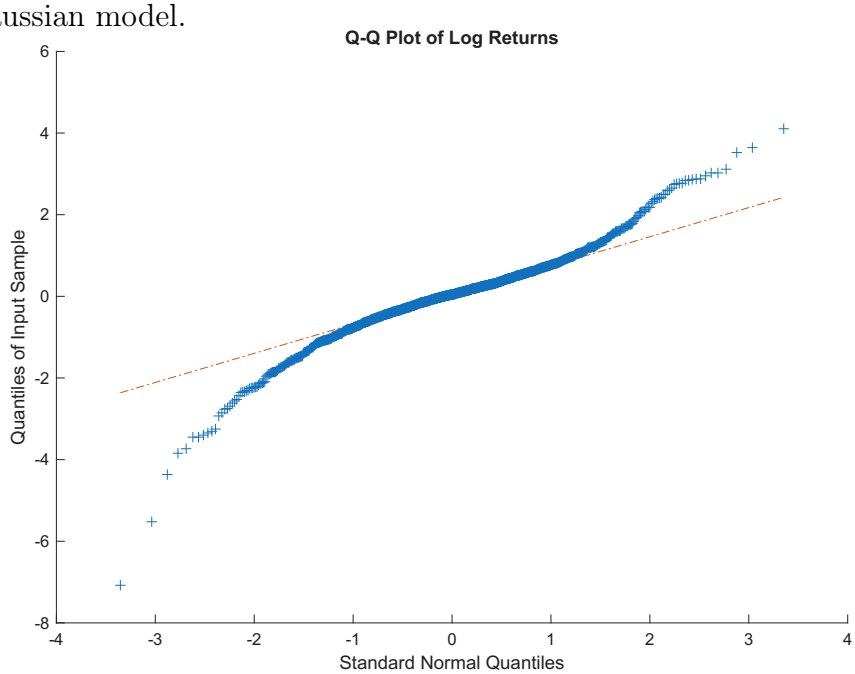
The test on the returns Y_t rejects the null hypothesis, providing strong evidence against normality.

```
    --- Lilliefors Test ---
P-Value: 0.00100
Result: Reject Null Hypothesis (h=1)
Conclusion: H0 is not supported
```

Q-Q plot

The Q–Q (quantile-quantile) plot is a graphical tool used to compare the empirical distribution of a dataset with a theoretical distribution, typically the normal (Gaussian) distribution. In a Q–Q plot, if the data follows the theoretical distribution, the points should lie approximately along the diagonal line.

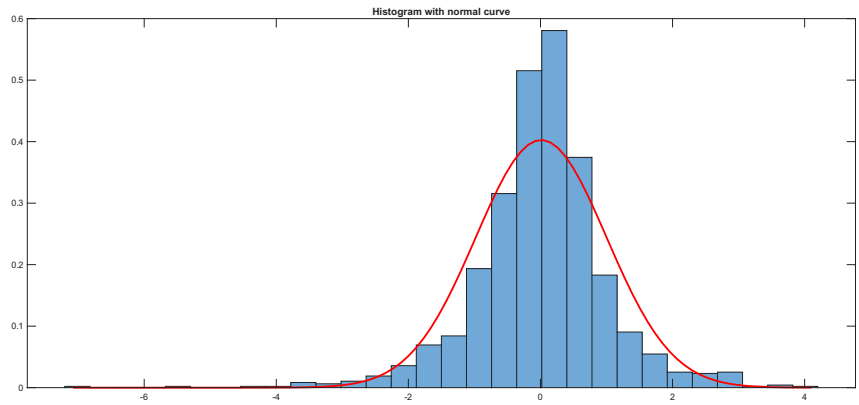
In our analysis, the Q–Q plot of the returns shows a S-shaped pattern, with deviations from the diagonal line. This indicates the presence of heavier tails compared to a normal distribution. Such deviations supports the failure of the Gaussian model.



Histogram

The graph displays the normal distribution with the mean and variance set to be equal to the empirical mean and variance of the data, and the histogram of the dataset.

From the plot, it is evident that the data exhibit heavy tails and deviate from the normal distribution. This indicates that extreme values occur more frequently than would be expected under a Gaussian model. Such behavior reinforces the conclusion that our data do not fit for the normality assumption.



Final considerations

Summarizing all the results obtained, we can conclude that the scaled log-returns lead to a definitive rejection of the **Independent and Identically Distributed (IID) Normal** assumption. In particular, the squared returns show strong autocorrelation, confirming the presence of **volatility clustering**, which implies that future risk depends on past market behavior. Moreover, the previous three tests performed indicate significant deviation from the Gaussian distribution. Thus, we cannot rely on the IID Gaussian model for this dataset. This suggests to adopt a dynamic volatility framework, such as the GARCH model, which will be deeply analized later in this report.

2 Problem 2

2.1 Theoretical introduction

Despite the rejection of IID Gaussian hypothesis, in this paragraph we assume that the model holds in order to analyze VaR and ES for this index. Before going through the computation, we will first give a theoretical background with definitions and formulas. The notation in the following lines are adopted in this way to be coherent with the instructions of the project.

Value-at-Risk (VaR) Definition

In financial risk management, **Value-at-Risk** (VaR) is one of the most widely used risk measure and it quantifies the potential loss in value of a risky asset or portfolio over a specific time period.
We define

$$X := V_1 - V_0 R_0$$

and the discounted loss as

$$L := -\frac{X}{R_0} = V_0 - \frac{V_1}{R_0}.$$

$\mathrm{VaR}_p(X)$ is defined in the following way

$$\begin{aligned}\mathrm{VaR}_p(X) &= \min\{m \in \mathbb{R} : \mathbb{P}(L > m) \leq p\} \\ &= \min\{m \in \mathbb{R} : F_L(m) \geq 1 - p\} \\ &= F_L^{-1}(1-p).\end{aligned}$$

So, $\mathrm{VaR_p}$ is a quantile of the loss distribution L. We remind that, given a random variable L, the α-quantile is the smallest number q_α such that

$$\mathbb{P}(L \leq q_\alpha) \geq \alpha.$$

Expected Shortfall (ES) Definition

While VaR provides a threshold loss value, it does not indicate the severity of losses beyond that threshold. **Expected Shortfall** (ES), also known as Conditional Value-at-Risk (CVaR), addresses this limitation by measuring the expected loss given that the loss exceeds the VaR threshold. Intuitively, Expected Shortfall measures the average magnitude of losses when conditional on the occurrence of extreme adverse outcomes.

Formally, for tail probability p:

$$\boxed{\mathrm{ES}_p(X) = \mathbb{E}[X \mid X > \mathrm{VaR}_p(X)]}.$$

An equivalent representation that is often more practical for calculations is the one that expresses ES as the average of VaR at level p:

$$\mathrm{ES}_p(X) = \frac{1}{p} \int_0^p \mathrm{VaR}_u(X)\, du.$$

Deriving VaR and ES formulas

Consider an investor holding one unit of an index fund with spot price V_{t-1} at time $t-1$. The net gain from $t-1$ to t is:

$$X = V_t - V_{t-1}.$$

We ignore interest rate effects for this short horizon. The scaled log returns are defined as:

$$Y_t = a\left(\log(V_t) - \log(V_{t-1})\right),$$

with $a = 100$, and are assumed to be independent and identically distributed (IID) with:

$$Y_t \sim N(\mu, \sigma^2).$$

Let $Z_t = \log(V_t/V_{t-1})$ be the log return. Then:

$$Y_t = aZ_t \quad \Rightarrow \quad Z_t = \frac{Y_t}{a}.$$

Since $Y_t \sim N(\mu, \sigma^2)$, we have:

$$Z_t \sim N\left(\frac{\mu}{a}, \frac{\sigma^2}{a^2}\right).$$

Denoting

$$m = \frac{\mu}{a}, \quad s = \frac{\sigma}{a},$$

the net gain can be written as

$$X = V_{t-1}\left(e^{Z_t} - 1\right) = g(Z_t),$$

where $g(z) = V_{t-1}(e^z - 1)$.

Let us now derive formulas for $\mathrm{VaR}_p(V_t - V_{t-1})$ and $\mathrm{ES}_p(V_t - V_{t-1})$.

$\mathbf{VaR}_p$ We want now to compute the Value-at-Risk for $X = V_t - V_{t-1}$. Using the definition above, taking $R_0 = 1$, $L = -X$ and

$$\begin{aligned}\mathrm{VaR}_p(X) = \mathrm{VaR}_p\big(g(Z_t)\big) &= F^{-1}_{-g(Z_t)}(1-p) \\ &\overset{\text{prop. 6.4}}{=} F^{-1}_{-g(Z_t)}(p) \\ &\overset{\text{prop. 6.3}}{=} -g\big(F^{-1}_{Z_t}(p)\big)\,.\end{aligned}$$

Substituting with the function $g(z) = V_{t-1}(e^z - 1)$,

$$\mathrm{VaR}_p(X) = V_{t-1}\left(1 - e^{F^{-1}_{Z_t}(p)}\right).$$

Noting that if $Z \sim N(\mu, \sigma^2)$ then $F^{-1}_Z(x) = \mu + \sigma\Phi^{-1}(x)$. In our case, since $Z_t \sim N\left(\frac{\mu}{a}, \frac{\sigma^2}{a^2}\right)$,

$$\boxed{\mathrm{VaR}_p(X) = V_{t-1}\left[1 - \exp\left(\frac{\mu}{a} + \frac{\sigma}{a}\Phi^{-1}(p)\right)\right]}.$$

$\mathbf{ES}_p$ We can now proceed computing the Exspected Shortfall. By definition,

$$\mathrm{ES}_p(X) = \frac{1}{p}\int_0^p \mathrm{VaR}_u(X)\,du.$$

Substituting our VaR formula derived:

$$\mathrm{ES}_p(X) = V_{t-1}\left[1 - \frac{1}{p}\int_0^p \exp\left(m + s\Phi^{-1}(u)\right)du\right].$$

Substituting $m = \frac{\mu}{a}$ and $s = \frac{\sigma}{a}$, and evaluating the integral we obtain the final formula:

$$\boxed{\mathrm{ES}_p(X) = V_{t-1}\left[1 - \frac{1}{p}\exp\left(\frac{\mu}{a} + \frac{\sigma^2}{2a^2}\right)\Phi\left(\Phi^{-1}(p) - \frac{\sigma}{a}\right)\right]}.$$

2.2 Computation and observations

Following the theoretical derivation, we can proceed analyzing our case.

To evaluate the model's performance, we partition the dataset:

1. **Training Set:** The first 1,151 observations are used to estimate the model parameters: the mean $\hat{\mu}$ and the standard deviation $\hat{\sigma}$. These two parameters will be constant in the computation

2. **Test Set:** The last 100 observations serve as the evaluation period. For each day t in this set, we compute the risk measures $\mathrm{VaR}_p(X)$ and $\mathrm{ES}_p(X)$ using the parameters derived from the training set and the previous day's spot price V_{t-1}.

We then compare the VaR_p estimates with the realized daily losses, defined as

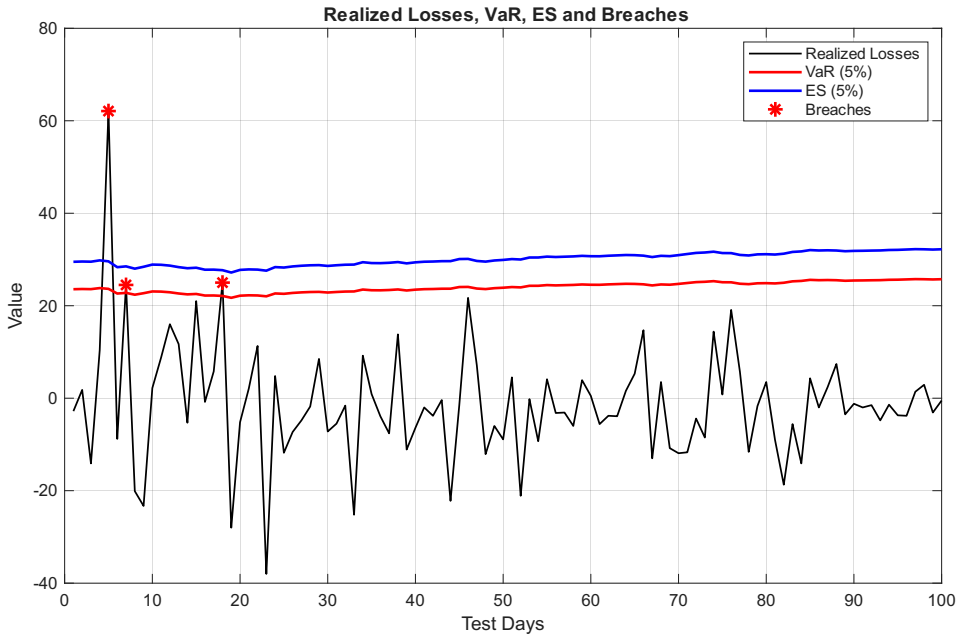
$$L_t = V_{t-1} - V_t.$$

Finally, we compute the number of breaches, defined by

$$V_{t-1} - V_t > \mathrm{VaR}_p,$$

that is, when the observed loss exceeds the Value at Risk at confidence level p.

Plotting all in the same graph, for $p = 0.05$ we obtain the following:



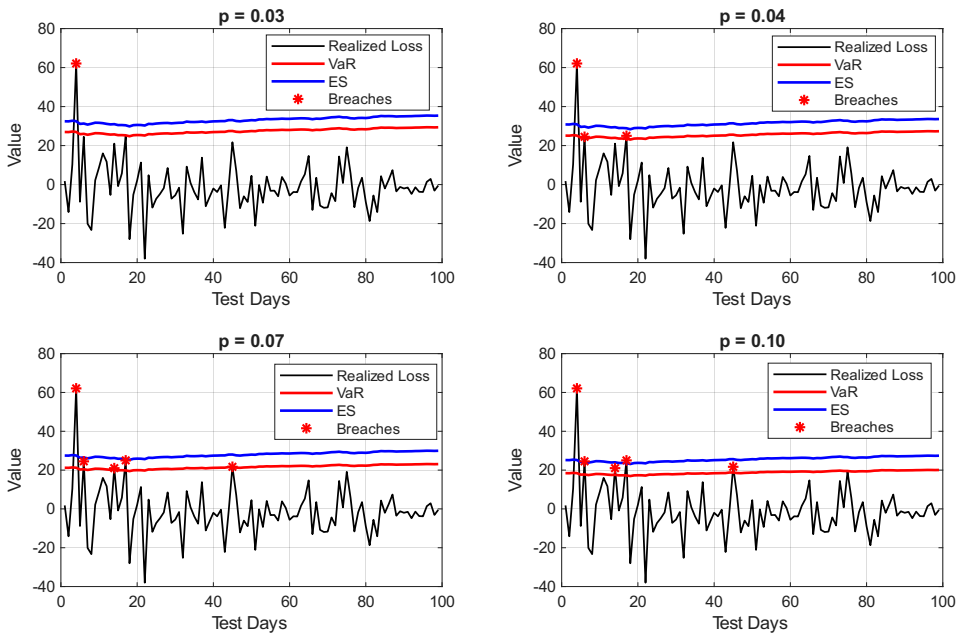
First, it is possible to note that VaR and ES are represented by an almost flat line, since the parameters are constant. For a significance level of $p = 0.05$ over a test set of $n = 100$ days, the expected number of breaches is simply:

$$n \times p = 100 \times 0.05 = 5.$$

However, our empirical analysis of the test set resulted in only **3 breaches**.

The fact that the observed breaches (3) are fewer than expected (5) indicates that the IID Normal model was over-conservative during the test period, setting the VaR threshold too high.

Let us now compute the analysis changing the value of p.



Iterating the process for different values of p leads us to same conclusions. For example, when $p = 0.03$, only 1 breach occurs compared to 3 expected, and for $p = 0.1$ we only obtain 5 breaches. We can therefore conclude that the IID model is too safe and overestimates the risk. From a financial perspective, this could lead the investor to set aside more capital than necessary to cover potential losses, reducing the capital available for profitable investment.

In the following steps, by introducing new models, we expect to capture better the risk for our index.

3 Problem 3

3.1 GARCH processes

A Generalized Autoregressive Conditional Heteroskedasticity (GARCH) model is a widely used time–series model designed to describe and forecast time–varying volatility. Unlike models with constant variance, GARCH models allow the conditional variance of the process to change over time, capturing important empirical features of financial returns such as volatility clustering.

Let $\{r_t\}$ denote a return series. A GARCH(p, q) model is defined as

$$r_t = \mu + \varepsilon_t, \qquad \varepsilon_t = \sigma_t z_t,$$

where $\{z_t\}$ is a sequence of i.i.d. random variables with zero mean and unit variance, and the conditional variance σ_t^2 evolves according to

$$\sigma_t^2 = \alpha_0 + \sum_{i=1}^{p} \alpha_i \varepsilon_{t-i}^2 + \sum_{j=1}^{q} \beta_j \sigma_{t-j}^2.$$

Here, p represents the number of lagged squared innovations (ARCH terms) and q the number of lagged conditional variances (GARCH terms).

3.2 Computation

The objective is to investigate how the VaR_p changes when the returns are modeled using a GARCH process. In order to estimate the GARCH model,

it is necessary to determine the appropriate orders p and q. The selection of (p, q) can be done by minimizing the Bayesian Information Criterion (BIC), defined as

$$\mathrm{BIC} = -2 \log L + m \log(n),$$

where:

- $\log L$ is the log-likelihood of the fitted model;
- m is the number of parameters estimated by the model;
- n is the number of observations in the training dataset.

The computation

We now briefly describe the procedure used to estimate $\hat{p}$ and $\hat{q}$.

1. We consider the training dataset consisting of the first 1151 observations. The empirical mean is computed and subtracted from the data. We center the data in order to focus the estimation on the conditional variance;
2. We estimate the orders p and q by generating a grid of candidate values,

   $$p, q \in \{1, 2, 3, 4, 5\}.$$

   For each pair (p, q), a GARCH(p, q) model is fitted to the data using maximum likelihood estimation. In particular, for each specification:

   - the sample size is set to $n = 1151$;
   - the number of estimated parameters is $m = 1 + p + q$;
   - the model parameters are estimated by maximizing the log-likelihood function;
   - the Bayesian Information Criterion (BIC) is computed.
3. Finally, we select the values $\hat{p}$ and $\hat{q}$ that minimize the BIC.

The results of the estimation show that the minimum BIC is obtained for

$$\hat{p} = 1, \qquad \hat{q} = 1,$$

indicating that a GARCH$(1, 1)$ model provides the best trade-off between goodness of fit and model complexity.

4 Problem 4

In the previous section, we have derived the optimal values $\hat{p}$ and $\hat{q}$ for the GARCH model. We will now briefly show how to derive the standardized residuals

$$\frac{Y_t - \hat{\mu}}{\hat{\sigma}_t}$$

and then we proceed with the diagnostic checks.
In order to compute the standardized residuals, we first estimate $\hat{\mu}$ and $\hat{\sigma}_t$ using the training set. MATLAB provides several useful functions; the following code was used in our analysis:

```
model = garch(pOpt, qOpt);
[fit, ~, logL] = estimate(model, Y_train, 'Display', 'off');
condVar = infer(fit, Y_train);
```

The fit output is the following:

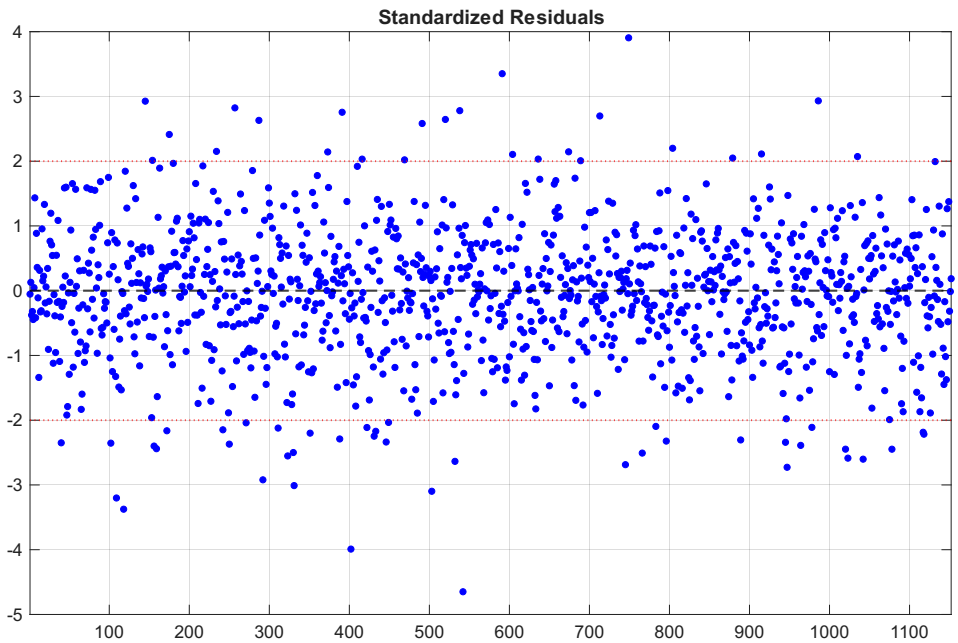
```
fit =
  garch with properties:

   Description: "GARCH(1,1) Conditional Variance Model
                 with Offset (Gaussian Distribution)"
    SeriesName: "Y"
  Distribution: Name = "Gaussian"
             P: 1
             Q: 1
      Constant: 0.0182611
         GARCH: {0.845755} at lag [1]
          ARCH: {0.143317} at lag [1]
        Offset: 0.0459154
```

This is the set of all the parameters that appear in the GARCH theoretical definition, noting that we have only one ARCH value α and one GARCH value β as the orders p and q are both 1. The parameter "Offset" coincides with $\hat{\mu}$.

Moreover, by taking the square root of `condVar` we obtain $\hat{\sigma}_t$; therefore, we are able to compute the standardized residuals and analyze them.

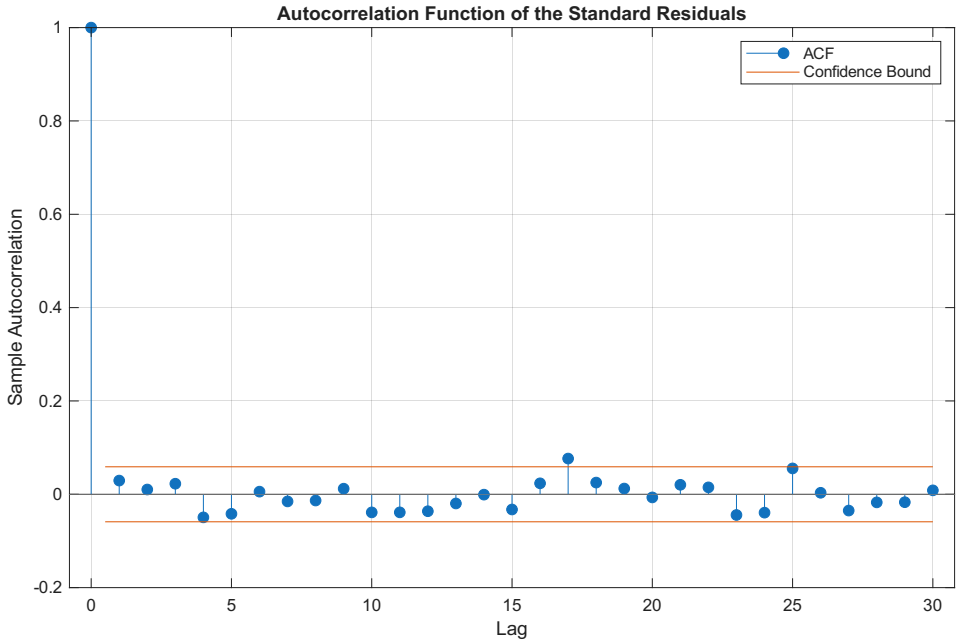
Standardized Residuals plot

The following figure shows the plot of the standardized residuals. The residuals fluctuate around zero and do not exhibit any visible trends or systematic patterns. This behavior suggests that the residuals are approximately random, providing evidence that the model adequately captures the time-varying volatility in the data.



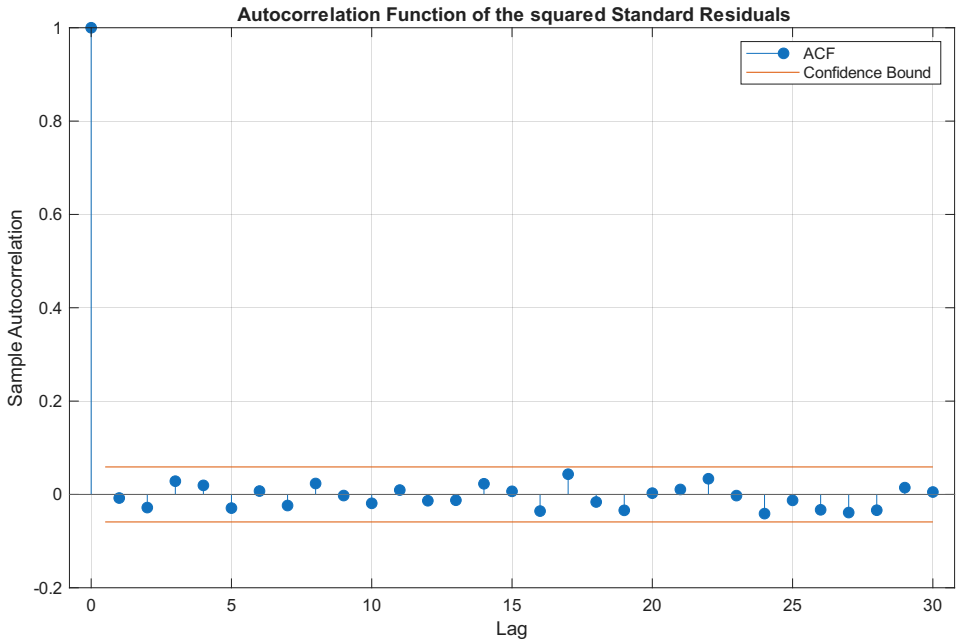
ACF of Standardized Residuals

The ACF plot of the standardized residuals, using 30 lags, shows that most autocorrelations lie within the confidence bounds. This indicates that the standardized residuals are approximately uncorrelated. This is a positive sign for the selected GARCH model, suggesting that it has successfully captured the time-varying nature of volatility.



ACF of squared Standardized Residuals

By analyzing the autocorrelation function of squared Standardized Residuals, we observe that almost all the values lie inside the confidence bound, hence we have no signs of correlation; the conclusion is the same stated above, showing a good model fit.



Lilliefors Test for Normality

The hypotheses of the test are as follows:

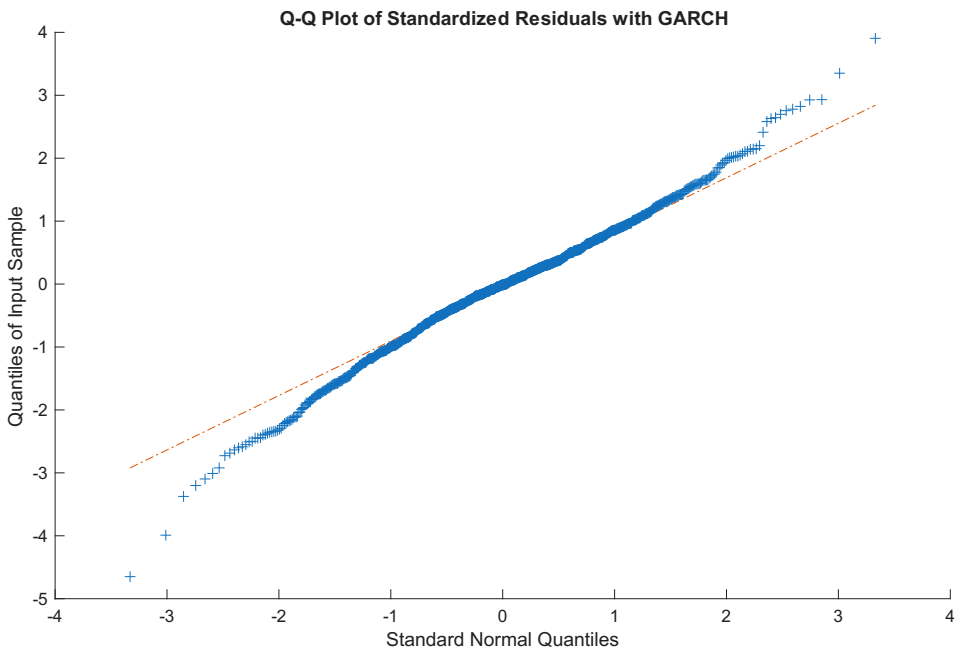
- H_0: The data follow a normal distribution;
- H_1: The data are not normally distributed.

The test on the Standardized Residuals rejects the null hypothesis, providing strong evidence against normality.

```
--- Lilliefors Test ---
P-Value: 0.00100
Result: Reject Null Hypothesis (h=1)
Conclusion: Significant evidence AGAINST the assumption.
```

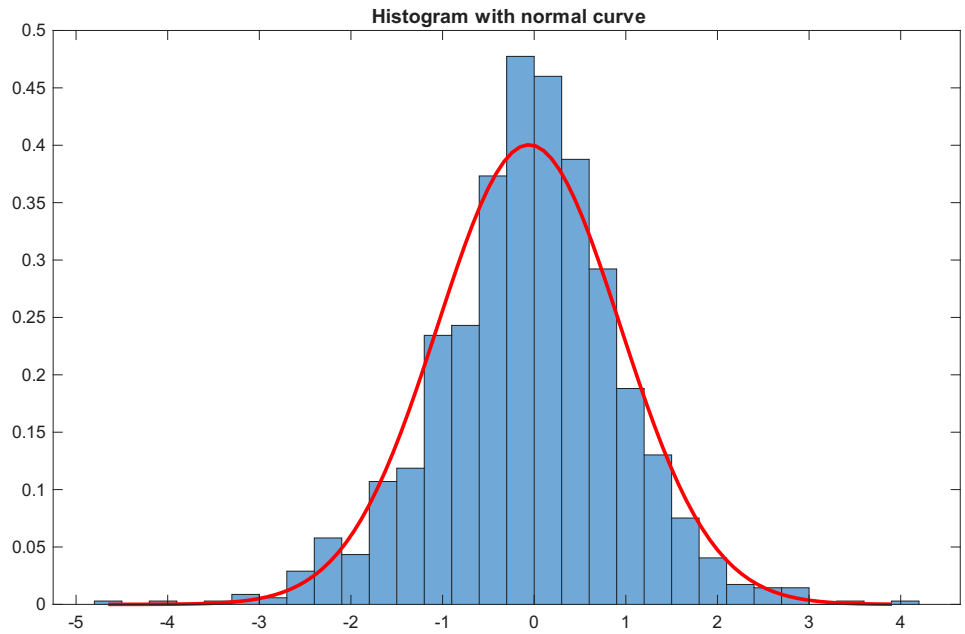
Q–Q Plot

The Q–Q plot of the Standardized Returns shows a S-shaped pattern, with deviations from the diagonal line. This indicates the presence of heavier tails compared to a normal distribution. Such deviations support the fact that the Standardized Residuals are not Gaussian.



Histogram

Finally, also the histogram shows that extreme values occur more frequently than would be expected under a Gaussian model. Such behavior reinforces the conclusion that the Standardized Returns do not fit a Gaussian distribution.



Final considerations

From these diagnostic tests, we can derive two main conclusions regarding the model's performance. First, the GARCH tool offers a substantial improvement over the IID Gaussian model which comes from the nature of the models: while the IID framework wrongly assumes homoskedasticity, meaning constant variance over time, the GARCH model is made to handle changing variance (heteroskedasticity). By allowing the variance $\hat{\sigma}_t^2$ to evolve in time, the GARCH model successfully captures the market's dynamic. This is pointed out by the standardized residuals: when we normalize the returns by the estimated volatility ($\frac{Y_t}{\hat{\sigma}_t}$), the volatility clustering disappears, hence the residuals behave like white noise, indicating that the variance dynamics have been correctly filtered out. On the other hand, the normality tests reveal that the standardized residuals still deviate from the Gaussian assumption exhibiting "fat tails". This result is consistent with the nature of our log-returns: while GARCH explains the clustering of volatility,it is not able to capture the frequency of extreme market events.

5 Problem 5

The objective of this section is to compute VaR and ES using the GARCH(1,1) model, providing a comparison to the results obtained in Section 2.

The idea is to forecast the next day's log-return mean and variance (volatility) using the fitted GARCH model. Then compute the $\text{VaR}_{0.05}$ and $\text{ES}_{0.05}$ for the first observation in the test set. For each subsequent point in the test set, progressively add all previous observations and refit the GARCH model. The $\text{VaR}_{0.05}$ and $\text{ES}_{0.05}$ are then reforecast at each new time step.

Specifically, we proceeded in the following way:

1. We first used the `estimate` function to train the model on the available history data, (starting from our training set) that will be updated in each iteration.

```
[estModel, ~, ~] = estimate(model, Y_history, 'Display','off');
```

2. we used the `forecast` function to predict the conditional variance ($\hat{\sigma}^2_{t+1}$) for the next day, while the conditional mean ($\hat{\mu}$) is obtained from the model's `Offset` property.

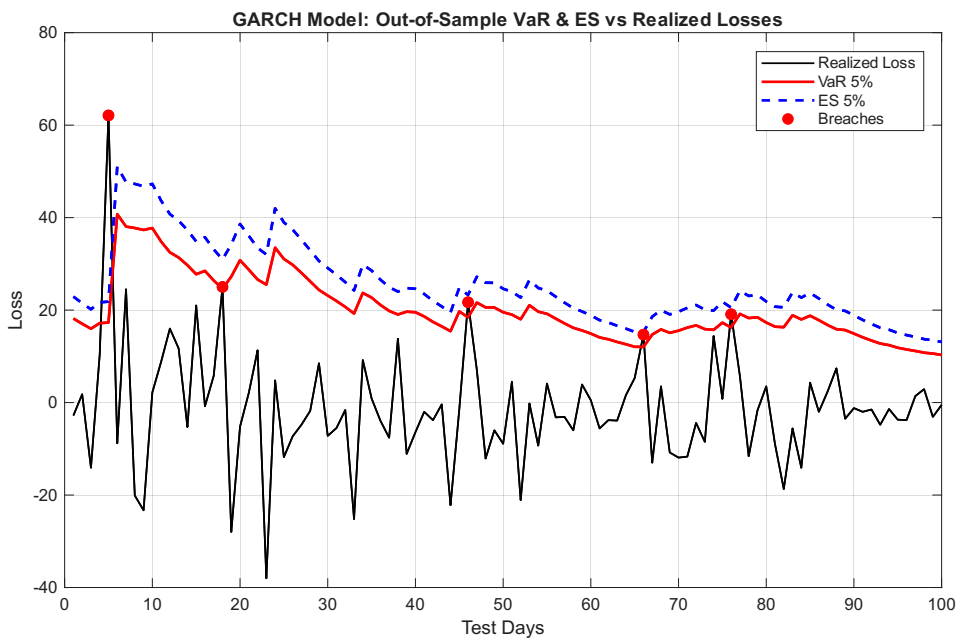
```
vF = forecast(estModel, 1, 'Y0', Y_history);
sigmaF = sqrt(vF);      % predicted volatility for tomorrow
muF    = estModel.Offset; % predicted mean (Offset)
```

3. We used the parameters just obtained to compute VaR and ES formulas derived in Problem 2.

```
VaRF = 1 - exp( (muF+ sigmaF * norminv(p, 0, 1)) / a);
ESF  = 1 - (1/p) * exp( muF/a + (sigmaF^2)/(2*a^2) ) ...
          * normcdf( norminv(p, 0, 1) - (sigmaF/a) );
VaR(t) = V_test_prev(t) * VaRF;
ES(t) = V_test_prev(t) * ESF;
```

4. We updated $Y_{history}$ with current return and repeated the loop.

```
Y_history = [Y_history; Y_test(t)];
```



The plot shows significant results. Firstly, we observe that the number of breaches matches the expected theoretical value (5 breaches for a 5% confidence level). Although, trying the computation with $p = 0.1$ we observed only 6 breaches compared to the expected 10 ones; this shows that the risk is not perfectly modeled, and the case with 5 breaches might be a statistical coincidence. This could be caused by the fact that our dataset test is

made of 100 values, which could be considered small. However, this represents an improvement compared to the result obtained with the IID model. The most important aspect to point out is how the shape of the risk measures has changed. In the IID assumption, the VaR appeared as a flat line, due to constant parameters. In contrast, the GARCH-based VaR varies significantly, since the parameters change through the time, demonstrating that the model effectively adapts to market dynamics. For instance, when the market experiences large losses (volatility spikes), the VaR line immediately rises, to create a larger safety bound, as we can see in the first time interval. Conversely, when the market stabilizes, the VaR tightens again. This "reactivity" means that the risk measure adjusts to market conditions: when markets become volatile or unstable, the model quickly signals higher risk, so more capital is set aside for protection. When markets are calm, the perceived risk decreases, and the investor does not need to hold unnecessarily large capital. In this way, capital is used efficiently.

6 Problem 6

The residual diagnostics presented in Section 4 indicate that the standardized residuals do not follow a Gaussian distribution and exhibit a "fat-tailed" behavior. This suggests that the choice of the Gaussian GARCH model may be incorrect and motivates the use of a Student's t distribution to model the noise term z_t, as it provides a more appropriate characterization of extreme observations.

Following the same process adopted in Section 5, we compute the VaR by estimating a GARCH(1,1) model under the assumption of a Student's t distribution.

The following plot compares the VaR obtained under the Gaussian assumption (Section 5) and Student's t assumption.

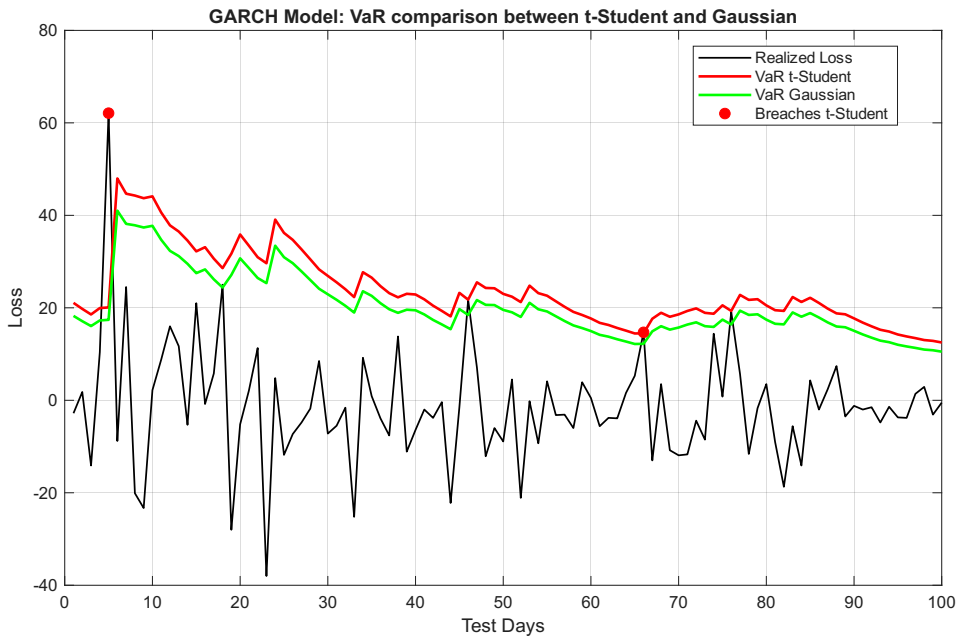


Figure 6.1: $p = 0.05$

First, we notice that the number of breaches obtained is less than the theoretical number. What's more, t-Student VaR is higher than the Gaussian VaR. This means that t-Student is overestimating the risk. This is consistent with the fact that we are including fat tails because t-Student distribution can capture potential extreme events. This is why the forecasted VaR becomes higher. To conclude, during the test period there weren't enough extreme losses to hit this higher treshold.

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